

Jiaxin Chen

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EDUCATION

Ph.D. in Statistics, The Ohio State University, Columbus, OH

08/2021 – 05/2027(*expected*)

Advisor: Xiaoxuan Cai

B.S. in Statistics, Xiamen University, China

08/2016 – 06/2020

GPA: 3.88/4.00 (1/40)

CORE COURSES

Machine Learning, Causal Inference, Bayesian Statistics, Stochastic Processes, Data Mining, Econometrics, Time Series Analysis

TECHNICAL SKILLS

- Programming Languages: Python, R, Rcpp
- Tools: Big Data (PySpark); Database (SQL); Machine Learning Frameworks (Scikit-Learn, PyTorch); Data Manipulation and Visualization (NumPy, Pandas, Matplotlib, dplyr, tidyr, ggplot2)

WORK EXPERIENCE

Data Science Internship

05/2025 – 08/2025

Ads Experimentation, Pinterest

- Designed and executed large-scale simulation studies to evaluate methods for **zero-inflated metrics**, comparing zero-inflated models with block Jackknife t-tests.
- Explored and benchmarked approaches for **ratio metrics**, including several Controlled-experiment Using Pre-Experiment Data (**CUPED**) and linearization methods, using A/A and A/B tests to assess FPR, accuracy, and minimum detectable effect (MDE).
- Optimized large-scale simulation pipelines through vectorization and parallel processing with **PySpark**, accelerating experimentation workflows and enabling more extensive analyses.

Graduate Research Associate

08/2024 – Present

Department of Statistics at The Ohio State University

- Collaborated with faculty in Statistics and Psychiatry, developing strong interdisciplinary research skills.

Graduate Teaching Associate

08/2021 – 05/2024

Department of Statistics at The Ohio State University

- Led weekly recitations and tutoring sessions for undergraduate students.

RESEARCH EXPERIENCE

Bayesian Modeling and Online Learning of Personalized Time-Varying Treatment Effects in mHealth

Supervised by Dr. Xiaoxuan Cai and Dr. Oksana Chkrebtii

02/2025 – Present

- Developed a **hierarchical Bayesian state-space model** for nonstationary mobile health data to estimate individualized, time-varying treatment effects and classify participants into clinically interpretable response profiles (harmful, neutral, and beneficial).
- Designed and implemented a **partially collapsed hybrid MCMC sampler** combining Gibbs sampling, Metropolis-Hastings updates, Forward Filtering Backward Sampling (FFBS), and Reversible-Jump MCMC for posterior inference and subgroup classification.
- Built posterior forecasting and online learning procedure based on sequential Monte Carlo to allow adaptive updating for new patients.
- Validated the framework through simulation studies and applied it to the Advancing Resilience and Coping study to evaluate Crisis Response Plan use and suicidal ideation risk.

Comparing the Impacts of Crisis Response Plan and Self-Administered Safety Plan Use on Clinical Outcomes

Supervised by Dr. Xiaoxuan Cai and Dr. Melanie Bozzay

Accepted by Suicide and Life-Threatening Behavior, 09/2025

- Conducted statistical modeling using Generalized Linear Mixed-Effects Models (**GLMM**) on ecological momentary assessment data.
- Examined how plan use frequency moderated associations with suicidal ideation, positive affect, and negative affect across treatment groups (Crisis Response Plan vs. Self-Administered Safety Plan).
- Found that more frequent CRP use is linked with lower risk of suicidal ideation and greater positive affect compared to the self-guided SP, with implications for clinical practice.

Prediction and Analysis of Bipolar Patients' Loneliness Index

03/2024 – 08/2024

Supervised by Dr. Xiaoxuan Cai

- Developed and optimized machine learning models (**MLP**, **XGBoost**, **Random Forest**, etc.), achieving an AUC of 70%+ for predicting loneliness. Implemented **LSTM RNN** for time series prediction, reaching over 80% accuracy and 90% AUC.
- Addressed missing data with moving averages and Time-aware LSTM. Applied feature selection and reduction techniques (univariate feature selection, recursive feature elimination, **SHAP**, and **Hierarchical PCA**) to enhance model performance.
- Analyzed the impact of predictors on loneliness using model-based feature importance, permutation importance, and SHAP values.

SELECTED AWARDS

Whitney Best Research Associate Award at the Ohio State University

2025

Lifang Scholarship at Xiamen University (top 2% at Xiamen University)

2019

University-level Merit Student (top 2% at Xiamen University)

2017